

Detector Ranking is Not Architecture-Invariant under Medical Covariate Shift

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Abstract

Post-hoc out-of-distribution (OOD) detectors are almost always ranked on a single backbone. Holding a medical covariate split fixed, that ranking is not architecture-invariant. On Camelyon17 hospital-2 (WILDS), Kendall's $W = 0.694$ across eight CNN families and seven scores (seed 42). MSP AUROC spans 0.515 (ResNet-50) to 0.883 (DenseNet-121); the DenseNet jump versus the ResNet mean is seed-robust (0.265 ± 0.065 , 5/5 seeds ≥ 0.15). ConvNeXt-Tiny and EfficientNetV2-S@224 hold on 4/5 seeds, MobileNetV3 and RegNetY on 3/5; EfficientNet-B3's seed-42 jump is not seed-stable (2/5). Feature-space scores (Mahalanobis, k NN) occupy ranks 1–2 on every official backbone. The same eight-family protocol on skin ISIC→PAD does not reproduce the MSP jump ($W = 0.791$; Maha rank-1 on 8/8). AUROC rank and OOD-triage coverage@risk 10% disagree on Camelyon and on MIDOG, and the sign of that disagreement flips. A pre-registered ViT-B/16 Camelyon MSP of 0.884 met a locked threshold of 0.6947. We introduce no new detector: do not copy an OOD ranking across backbones, and do not pick a backbone for triage from an AUROC table.

1. Introduction

On the Camelyon17 hospital-2 split, with the same seven post-hoc scores, ResNet-18/50 MSP AUROC is 0.574/0.515 while DenseNet-121 is 0.883 and EfficientNet-B3 (seed 42) is 0.803 (Fig. 1). A practitioner who copies an OpenMIBOOD-style ResNet-50 recipe on this shift and one who uses DenseNet are not looking at the same detector ranking.

That observation is *not* a claim that detector ranking was previously assumed architecture-invariant. On natural-image semantic OOD, ranking already moves with architecture [2, 3, 23, 31]. OpenMIBOOD [8] already showed that ImageNet method ranking does not transfer to medical data and that feature-space scores outperform logit scores un-

der a frozen classifier per domain. Our question is orthogonal: **holding a medical covariate-shift dataset fixed, how much does detector ranking move when the backbone changes?**

We measure that movement at zoo scale on Camelyon17, show that it does not replicate as an MSP jump on skin ISIC→PAD, and show that AUROC versus operational coverage@risk 10% disagrees with a *domain-dependent sign*. We pre-registered a ViT-B/16 MSP threshold before training that model. We do not claim a new detector, a Maha>MSP discovery, or a two-domain MSP-jump law.

Contributions.

- Camelyon-17, zoo-scale concordance.** Official Kendall $W = 0.694$ ($n=8$, 7 methods, seed 42). The seed-robust MSP jump is DenseNet-121 versus ResNet (0.265 ± 0.065 , 5/5). ConvNeXt-Tiny and EfficientNetV2-S@224 are 4/5; MobileNetV3 and RegNetY are 3/5; EfficientNet-B3 is 2/5. Matching the four non-Adam zoo members to ResNet-Adam does not remove those seed-42 jumps. Feature-space scores stay in the top two ranks. This is not a claim that architecture-sensitive ranking is new.
- Domain asymmetry.** Skin ISIC→PAD ($n=8$, $W=0.791$): Maha rank-1 and k NN rank-2 on all eight; MSP stays in a narrow band without a Camelyon-style ResNet-vs-rest jump.
- Utility mismatch is unpredictable.** Not that AUROC and coverage-based ranking can disagree (AURC/AUGRC already assume that [6, 26]). The *sign* of AUROC versus coverage@risk 10% disagreement flips between Camelyon and MIDOG, with no rule to predict which backbone wins.
- A locked prediction.** Before training: ViT-B/16 Camelyon MSP ≥ 0.6947 . Observed 0.884. Held out of the 8-CNN W on purpose.

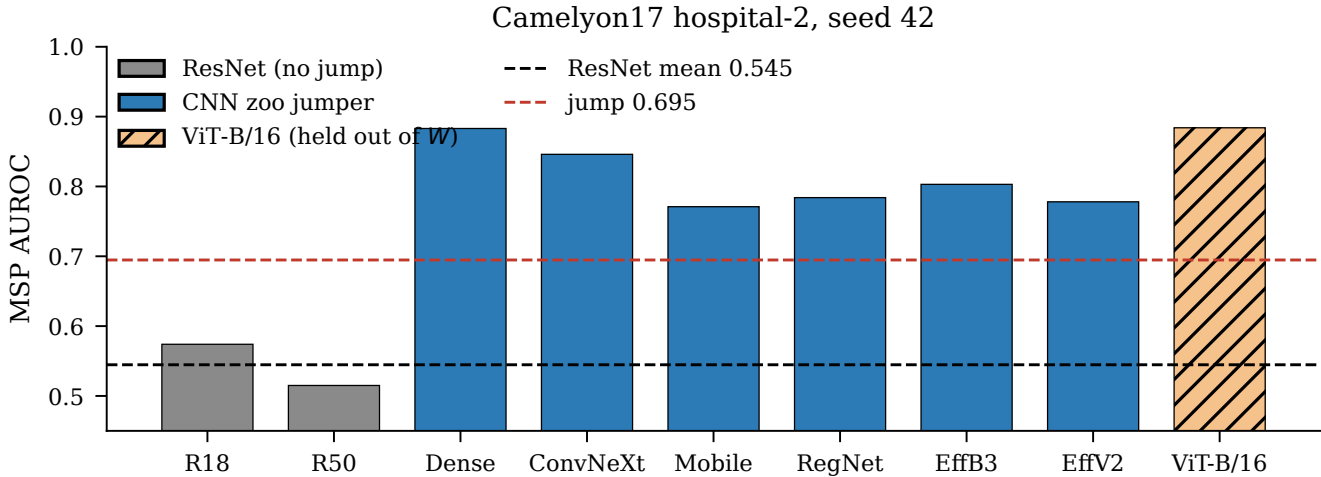


Figure 1. **Camelyon hospital-2 MSP AUROC, seed 42.** Dashed lines: frozen ResNet mean (0.545) and jump threshold (0.695). ViT-B/16 was pre-registered and is not a ninth column of Kendall W . DenseNet, ConvNeXt, MobileNet, RegNet, EffB3, and EffV2-S@224 all clear the bar at this seed; ResNets do not.

2. Related work

OpenMIBOOD [8] ranks 24 post-hoc detectors on three medical benchmarks with a **single frozen classifier per domain** (ResNet-50 on MIDOG, ResNet-18 on PhaKIR, R(2+1)D on OASIS-3). It shows that ImageNet method ranking does not transfer to medical data and that feature-space scores outperform logit scores under that fixed-architecture protocol. It does not measure whether the detector ranking itself is stable across architectures. We keep their MIDOG split and public ResNet-50 checkpoint as an external reference, and ask the orthogonal question: holding the dataset fixed, how much does detector ranking move when the backbone changes?

OpenOOD v1 [29] *fixes architecture on purpose* (LeNet / ResNet-18 / ResNet-50). That design choice is about v1, not about the later literature.

Architecture-sensitive post-hoc ranking *has* been measured on natural images. OpenOOD v1.5 [31] evaluates ImageNet-1K post-hoc methods on ResNet-50, ViT-B/16, and Swin-T. Figure 3 of that paper reports that “different post-processor may favor different architecture”: ASH/ReAct degrade on transformers, while RMDS suits them. That study uses three models, semantic near/far OOD, and a qualitative comparison. It does not compute Kendall’s W , does not sweep eight CNN families, and is not a covariate hospital shift. It is the closest natural-image cousin; we do not treat v1.5 as an architecture-fixed protocol.

Szyc *et al.* [23] show that detector rankings change across three ResNet-101/110 CIFAR implementation variants with matched closed-set accuracy, and across random seeds, with feature-space methods especially unsta-

ble. That is a reliability result about uncontrolled experimental details on CIFAR semantic OOD, not an eight-family zoo that holds one medical covariate split fixed. Datko *et al.* [3] report a preferred post-hoc detector *per CNN/ViT line*, claimed to be stable across OOD benchmarks for that model—the opposite of a single-split concordance statistic. Claros Olivares and Brockmeier [2] compare a scratch CNN to a fine-tuned ViT and organize confidence-score rankings into cliques by shift severity on CIFAR / TinyImageNet. Their released results rank scores with AUROC/AUGRC; they do not report an AUROC-rank versus coverage@risk 10% disagreement whose sign flips across medical covariate domains.

What remains is not the existence of architecture-sensitive ranking. That existence is published on natural-image semantic OOD. This paper measures one medical covariate-shift split, held fixed; an eight CNN-family zoo plus a precommitted ViT; Kendall’s W per domain; a structured MSP jump whose seed-robust core is DenseNet versus ResNet, with feature methods in ranks 1–2; skin non-replication of the MSP jump; an iWildCam miss on natural-image covariate shift; and the coverage sign-flip below. iWildCam is the natural-image covariate check we actually ran; OpenOOD v1.5 Fig. 3 is semantic ImageNet, not a substitute.

OoD-Bench [30] quantifies OOD *generalization*, not post-hoc detection; we cite it as venue precedent for a two-axis evaluation paper. WILDS Camelyon17 and iWildCam [14] supply the hospital / camera-trap covariate splits; iWildCam is a negative bound, not a third main domain.

3. Setup

3.1. Shifts

Camelyon17 WILDS [14]: hospitals 0, 3, 4 as ID and hospital 2 as OOD; founding cell, official W , $n=8$. Skin ISIC→PAD [19]: feature rank-stability without an MSP jump. MIDOG [8]: OpenMIBOOD 1a vs. 1b+1c, coverage sign-flip; official W is $n=3$. We build on OpenMIBOOD’s public MIDOG split and ResNet-50 checkpoint as an external reference. Public R50 AUROC ~ 0.59 is *theirs*; in-house R50 MSP 0.512 is *ours*. We never mix the two.

3.2. Backbones

Eight CNN-family models, ImageNet initialization, 10-epoch fine-tune, seed 42, train-frac 1.0 on Camelyon: ResNet-18/50 [9], DenseNet-121 [12], ConvNeXt-Tiny [18], MobileNetV3-Large [11], RegNetY-3.2GF [20], EfficientNet-B3 [24] at native 300, and EfficientNetV2-S [25] at 224 (native 384 dropped). Standard fine-tune recipes (Adam / AdamW / SGD) follow each family’s common practice and are *not* forced-identical Adam. That con-found is measured in Sec. 4.3. Only resolution was isolated in a controlled pair (EffB3 @224 vs. @300; EffV2-S @224 vs. @384). ViT-B/16 [4] @224 is held out of W .

3.3. Scores and statistics

Seven post-hoc scores: MSP [10], Energy [17], energy of logit-normed logits (ELogitNorm), ViM [27], ReAct [21], Mahalanobis with Ledoit–Wolf covariance [15, 16], and k NN [22]. Primary concordance: **Kendall’s W per domain, never pooled**. Jump: $\text{MSP} - \text{mean}(\text{R18}, \text{R50}) \geq 0.15$. Feature drag: Maha or k NN on a new backbone below median($\text{R18}, \text{R50}, \text{EffB3}$) – 0.10. Official Camelyon ResNet mean $\text{MSP} = 0.544654$; jump iff $\text{MSP} \geq 0.695$ on that split.

Coverage@risk 10% is OOD-triage: rank OOD samples by MSP (higher = more ID-like) and take the largest coverage whose classifier risk is $\leq 10\%$. ID-selective Camelyon is vacuous (ID-val acc $\approx 99.5\%$) and is not cited.

We tested whether the Camelyon MSP pattern is an artifact of input resolution, temperature scaling [7], ID-only detector *selection*, a CE+SupCon recipe [13], or train-time HED stain jitter. None is a clean account (Sec. 5 and the supplement).

4. Results

4.1. Camelyon-17: official table and five-seed MSP

Official Kendall $W = 0.694$ ($n=8$, 7 methods). This is the only Camelyon number called “official W .” Leave-one-backbone recomputation of W ranges 0.676–0.738; that is *structural* sensitivity of the concordance statistic, not a 95% CI (supplement).

Feature methods occupy ranks 1–2 on every official backbone (Maha #1 on 6/8; k NN #1 on EffB3 and MobileNet). No backbone drags Maha/ k NN past the 0.10 feature-drag threshold. On Table 1, MSP spans 0.515–0.883; Maha 0.870–0.988; k NN 0.856–0.973. Incomplete concordance ($W=0.694$) is the logit ranks moving (Fig. 2).

Table 2 reports the full five-seed MSP matrix. DenseNet-121 is the confirmed seed-robust jump (5/5). ConvNeXt-Tiny and EfficientNetV2-S@224 are not seed-42 flukes (4/5); we do not promote them to 5/5. MobileNetV3 and RegNetY are 3/5. EfficientNet-B3 is 2/5: its seed-42 jump (+0.258) is a favorable draw relative to 0.142 ± 0.125 . The seed-42 matrix still shows a jump without feature drag. The prose claim is narrower than “6/6 non-ResNet, confirmed.”

4.2. Skin ISIC→PAD is not a second A

Official $W = 0.791$ ($n=8$). Maha is rank-1 and k NN rank-2 on **all eight** (Fig. 2). MSP sits in a narrow band (0.68–0.74). There is no Camelyon-style ResNet-vs-rest MSP jump. EffV2-S@224 MSP is 0.737; the native-384 cell that put k NN at rank 7 (MSP 0.823) is a resolution artifact and is not in official W . Skin is rank-stability of feature scores under a zoo: a replication of OpenMIBOOD’s feature>logit pattern *across architectures*, not an MSP-jump.

4.3. Resolution, then optimizer

Resolution is not the Camelyon DenseNet / seed-42 pattern. EffB3 Camelyon: 0.803 @300 $\rightarrow$ 0.787 @224 (the jump versus the official ResNet mean survives). EffV2-S Camelyon: 0.726 @384 $\rightarrow$ **0.778** @224 (the official cell; the jump is not a 384 artifact). On skin, the EffV2-S @384 k NN collapse *is* a resolution artifact and disappears at 224. These two facts are not a slogan that “resolution never matters.”

M1 — jumpers trained with ResNet Adam ($\text{lr}=10^{-4}$, $\text{wd}=10^{-4}$, batch 64). The four official zoo members whose standard recipe was not that Adam still jump versus the frozen official ResNet mean 0.545: ConvNeXt 0.820 (+0.275), MobileNetV3 0.737 (+0.193), RegNet 0.805 (+0.261), EffV2-S@224 0.873 (+0.328); all with id-val acc ≥ 0.994 . 4/4 clean hits. DenseNet and EffB3 already shared ResNet-Adam in the official zoo. Recipe mismatch does *not* explain the seed-42 jumps of those four members. M1 is seed 42 only; the five-seed MSP matrix is Table 2.

M2 — ResNets trained with ConvNeXt AdamW ($\text{wd}=0.05$). ResNet-18 MSP **0.773** (enters the 0.695 band; id-val 0.9957). ResNet-50 MSP 0.581 (stays out; id-val 0.9958). ResNet *can* jump under a non-default recipe. Architecture is not a clean variable, independent of optimizer. These two results point in opposite directions; we report both.

Backbone	input	Maha (rank)	kNN (rank)	MSP	seed-42 jump
ResNet-18	224	0.956 (1)	0.936 (2)	0.574	no
ResNet-50	224	0.896 (1)	0.866 (2)	0.515	no
DenseNet-121	224	0.988 (1)	0.973 (2)	0.883	yes
ConvNeXt-Tiny	224	0.928 (1)	0.916 (2)	0.846	yes
MobileNetV3-L	224	0.870 (2)	0.904 (1)	0.771	yes
RegNetY-3.2GF	224	0.964 (1)	0.956 (2)	0.784	yes
EfficientNet-B3	300	0.885 (2)	0.934 (1)	0.803	yes
EfficientNetV2-S	224	0.880 (1)	0.856 (2)	0.778	yes

Table 1. Camelyon hospital-2 AUROC, **seed 42** (official W). Jump iff $\text{MSP} \geq 0.6947$ vs. frozen ResNet mean 0.544654. MSP range is ResNet-50→DenseNet.

Backbone	42	43	44	45	46	jump mean $\pm$ sd	≥ 0.15
ResNet-18	0.574	0.712	0.601	0.524	0.581	—	—
ResNet-50	0.515	0.621	0.619	0.515	0.661	—	—
ResNet mean	0.545	0.667	0.610	0.520	0.621	—	—
DenseNet-121	0.883	0.865	0.868	0.845	0.826	0.265 $\pm$ 0.065	5/5
ConvNeXt-Tiny	0.846	0.799	0.803	0.764	0.785	0.207 $\pm$ 0.067	4/5
MobileNetV3-L	0.771	0.728	0.815	0.780	0.745	0.175 $\pm$ 0.081	3/5
RegNetY-3.2GF	0.784	0.778	0.836	0.830	0.724	0.198 $\pm$ 0.089	3/5
EfficientNet-B3	0.803	0.731	0.643	0.818	0.678	0.142 $\pm$ 0.125	2/5
EfficientNetV2-S@224	0.778	0.744	0.808	0.846	0.835	0.210 $\pm$ 0.089	4/5

Table 2. Same eight Camelyon hospital-2 MSP cells on all five seeds {42, 43, 44, 45, 46}. Jump = MSP minus that seed’s ResNet mean; bar 0.15. The last column is how many of those five seeds clear the bar. Official W is seed 42 only.

4.4. Held-out prediction: ViT-B/16

Locked *before* training, from the seed-42 zoo rule that non-ResNet-like cells jumped: $\text{MSP} \geq 0.6947$. Linear depth/parameter/receptive-field predictors are invalid on this $n=8$ and are not quoted. Observed MSP **0.884**, Maha 0.958, $k\text{NN}$ 0.941. The prediction holds. Not a ninth W column.

4.5. Coverage@risk: the sign flips

Protocol: OOD-triage coverage@risk 10% with MSP, not ID-selective coverage.

On Camelyon hospital-2, the locked $n=4$ utility table has MSP AUROC order EffB3@300 (0.803) > EffB3@224 (0.787) > ResNet-18 (0.574) > ResNet-50 (0.515), but coverage@risk 10% order ResNet-50 (0.149) > ResNet-18 (0.145) > EffB3@300 (0.107) > EffB3@224 (0.013). The AUROC winner is not the coverage winner (Fig. 3).

On MIDOG, official $n=3$ (in-house): MSP AUROC ResNet-50 0.512, EffB3 0.486, ResNet-18 0.438 (gaps are small). Coverage@risk 10%: EffB3 0.712, ResNet-18 0.442, ResNet-50 0.166. ResNet-50 is AUROC-highest and coverage-lowest. The permutation is *not* the Camelyon permutation.

We do not write “AUROC-winner = utility-loser” as a law, and we do not claim to have discovered that AUROC and coverage-based ranking can disagree—AURC/AUGRC

exist because that is already assumed. The increment is narrower: the *sign* of the disagreement flips between two medical covariate-shift domains under the same protocol, with no rule to predict which backbone wins. Do not choose a backbone for triage from an AUROC leaderboard; measure coverage@risk on the deployment domain. The Camelyon $n=8$ family span still disagrees (DenseNet AUROC 0.883 / coverage 0.078; MobileNet coverage 0.904); that is the same sign-flip claim, not a new official W .

4.6. Frozen foundation models

We do not claim a foundation-model invert on Camelyon. Linear-probe MSP: GigaPath [28] 0.664, Phikon [5] 0.605, UNI [1] 0.599, all >0.5. MIDOG UNI / GigaPath MSP 0.445 / 0.470 in a sign audit is a MIDOG-only footnote, not a Camelyon invert. FM Maha ~ 0.995 –1.0 is discussion support that feature>logit can survive a training-paradigm change, under the ranking-stability question, not a pillar.

5. Failed alternatives

Each of the following was a precommitted test. None is hidden.

Resolution (from Sec. 4.3). EffB3@224 MSP 0.787; EffV2-S@224 MSP 0.778. Not a pixel-size explanation of Table 1.

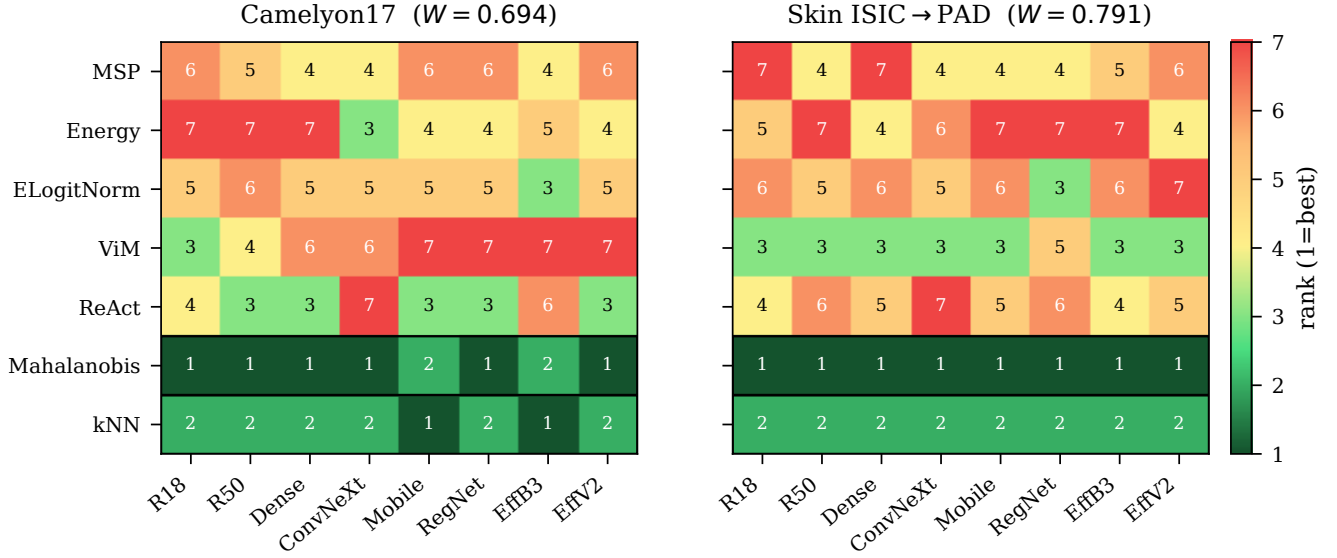


Figure 2. **Method rank by backbone** (1=best AUROC). Feature-space scores occupy ranks 1–2 on every official backbone in both domains. Logit ranks move on Camelyon ($W=0.694$) and much less on skin ($W=0.791$), where Maha is rank-1 on 8/8.

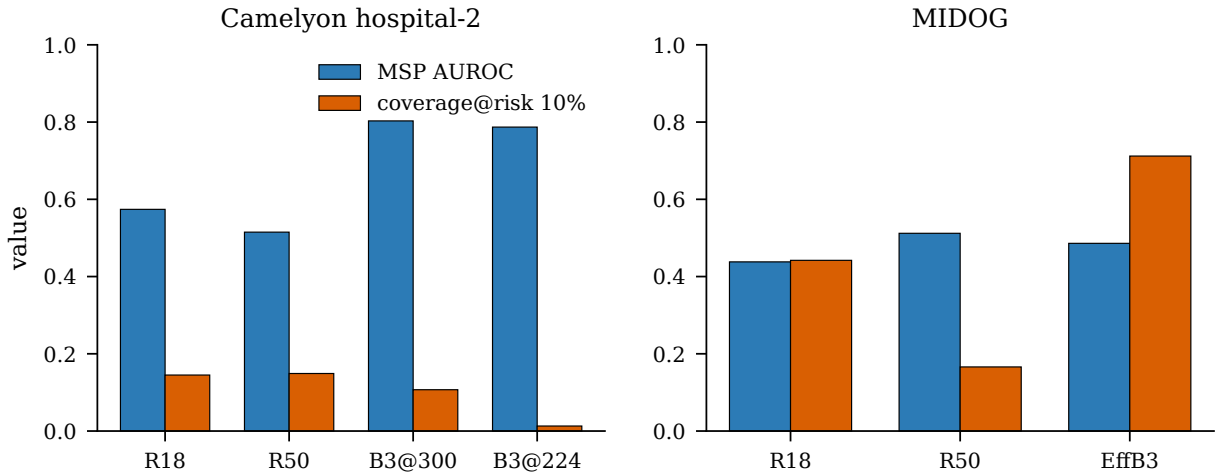


Figure 3. **AUROC vs. coverage@risk 10%**. On Camelyon the AUROC-high EfficientNets have the *lowest* triage coverage; on MIDOG the AUROC-high ResNet-50 is coverage-lowest. The sign of the disagreement flips.

Temperature T^* . Correlation of T^* with the MSP jump: $r=0.268$, $p=0.52$. DenseNet–ResNet-18 MSP gap $0.308 \rightarrow 0.308$ after T^* . Calibration/temperature is not the mechanism. We do not write a monotonicity theorem relating temperature scaling to MSP AUROC.

ID-only Maha/MSP gate. Selecting a score from ID-only statistics is a *selector*, not a *detector*. On Camelyon it picks Maha on 2/8 backbones; mean AUROC of the gated score 0.837 versus always-Maha 0.940. The bar was matching Maha and ViM. It does not.

CE+SupCon ($\lambda=0.1$, $\tau=0.07$), ResNet-18 + DenseNet. Gap $0.308 \rightarrow 0.238$ (23%; precommit bar was 50% / ≤ 0.154). ResNet-18 MSP $0.574 \rightarrow 0.515$

(worse; this is the SupCon cell, not official ResNet-50 MSP 0.515). Part of the shrink is the low model degrading. Dirty; not a recipe.

HED stain jitter, an in-house MIDOG $n=8$ zoo, and a shift-type extract (covariate / task / far-OOD) are in the supplement. Joint readout: no shift-type law, no stain mitigation, no Camelyon-style MSP jump on MIDOG. We do not open center-loss, triplet, a coverage surrogate, or an ensemble.

6. Discussion and limitations

Under this histopathology hospital shift, logit OOD scores move with the backbone: seed-robust for DenseNet (5/5), present on most seeds for ConvNeXt and EffV2-S@224 (4/5), weaker for MobileNet and RegNet (3/5), and seed-fragile for EfficientNet-B3 (2/5). Feature-space scores are the stable default *when the shift looks like Camelyon*. That is a protocol warning, not a new detector.

On skin, feature rank-stability holds *without* an MSP jump. Architecture-instability of MSP is not a cross-medical law. OpenMIBOOD’s feature>logit finding is replicated here under a zoo; the increment is ranking movement, when it happens, plus the operational AUROC↔coverage mismatch. ViT-B/16 was a locked prediction, not a post-hoc ninth CNN. M1 and M2 disagree about how cleanly that movement is “architecture” versus optimizer; both stand.

Limitations. (1) **iWildCam bound.** Locked 3-backbone cell: DenseNet MSP 0.575 vs. ResNet mean 0.604, jump -0.029 . Extra five CNNs vs. that domain’s ResNet mean: ConvNeXt 0.601, MobileNet 0.558, RegNet 0.580, EffB3 0.577, EffV2-S 0.650; 0/5 jump. We do not see the Camelyon MSP-jump on this non-medical covariate shift. The phenomenon may be biomedical-specific. We did not hunt a fourth medical domain and do not pool iWildCam with Camelyon W . (2) **No clean mitigation.** CE+SupCon is dirty (ResNet-18 MSP worse). HED stain jitter is dirty (DenseNet MSP $0.780 < 0.80$ precommit floor). The ID-only gate is a selector. (3) **MIDOG official W is $n=3$** (in-house R18/R50/EffB3, $W=0.857$). An in-house $n=8$ zoo shows no MSP jump and does not replace official W . (4) W is per-domain; do not average $0.694 / 0.791 / 0.857$. (5) No reader study: the operational claim stops at coverage@risk. (6) Architecture is confounded with its standard training recipe. 5/5 seed-robustness is DenseNet only. Official W stays seed 42. We did not isolate staining vs. scanner vs. tissue prep inside the hospital shift.

We will release the architecture-zoo evaluation and coverage@risk code, crediting OpenMIBOOD for MIDOG imglists, the public ResNet-50 checkpoint, and the 1a / cs-ID / near-OOD protocol.

7. Conclusion

Under Camelyon hospital shift, post-hoc detector choice is backbone-dependent: a seed-robust DenseNet MSP jump, weaker but directionally present jumps on ConvNeXt/EffV2 (4/5) and MobileNet/RegNet (3/5), and a seed-fragile EfficientNet-B3 cell. Feature-space ranks stay in the top two whether that jump is present (Camelyon) or absent (skin). A pre-registered ViT hit shows the Camelyon logit instability is not only a CNN-zoo accident; it did not appear

on iWildCam. AUROC and coverage@risk can rank backbones in opposite orders, and which way the disagreement points depends on the domain. There is no new method in this paper. The result is the protocol: do not copy an OOD ranking across backbones, and do not pick a backbone for triage from an AUROC table.

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